

AMAZON PRODUCT DASHBOARD

AN INTERACTIVE DATA VISUALIZATION USING R & SHINY

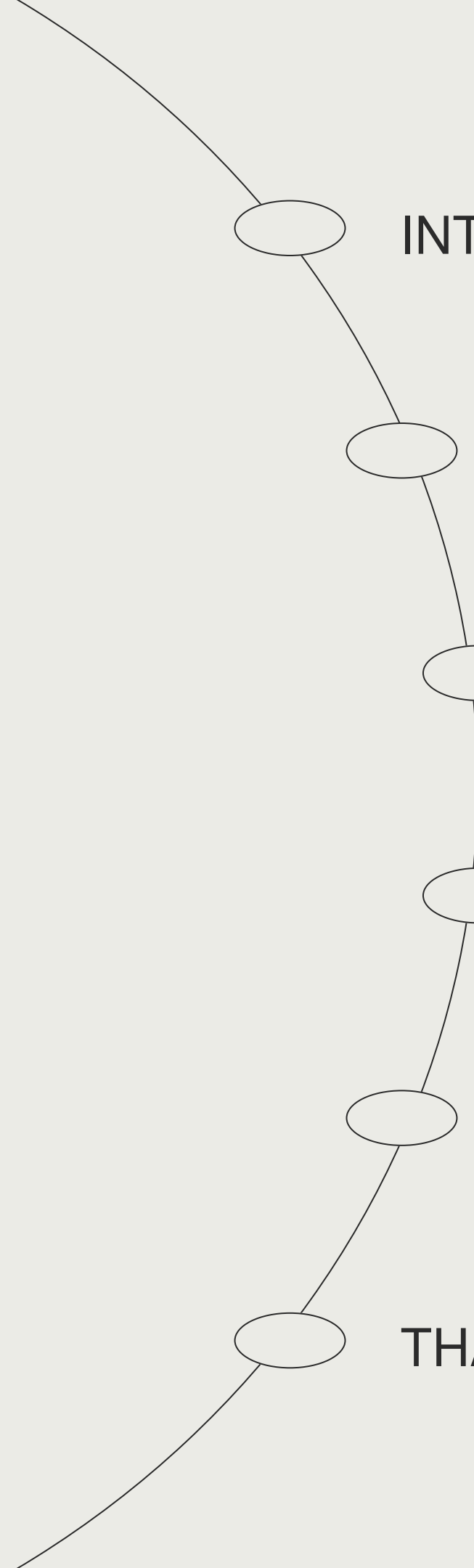
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SUPERVISOR : DR. YASIR HAFEEZ

DATE : 04/07/2025



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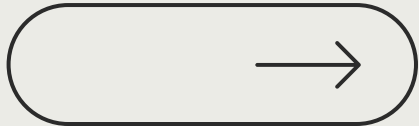
ABOUT THE DATA

TOOLS AND TECHNOLOGIES

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ACHIEVEMENTS & FUTURE
SCOPE

THANK YOU



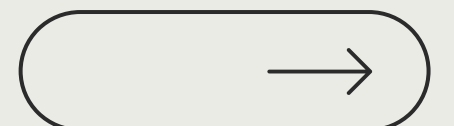
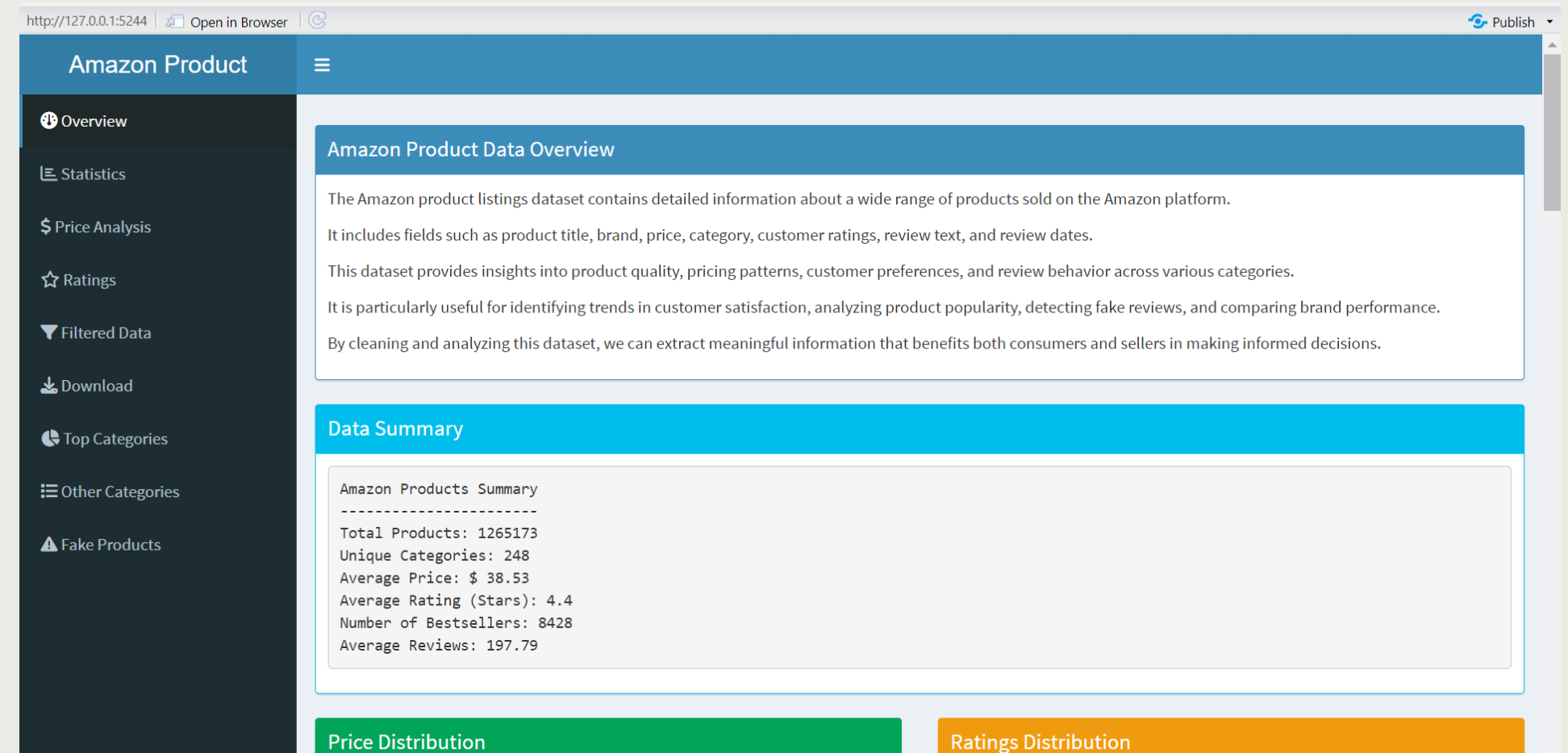
INTRODUCTION

Why This Project?

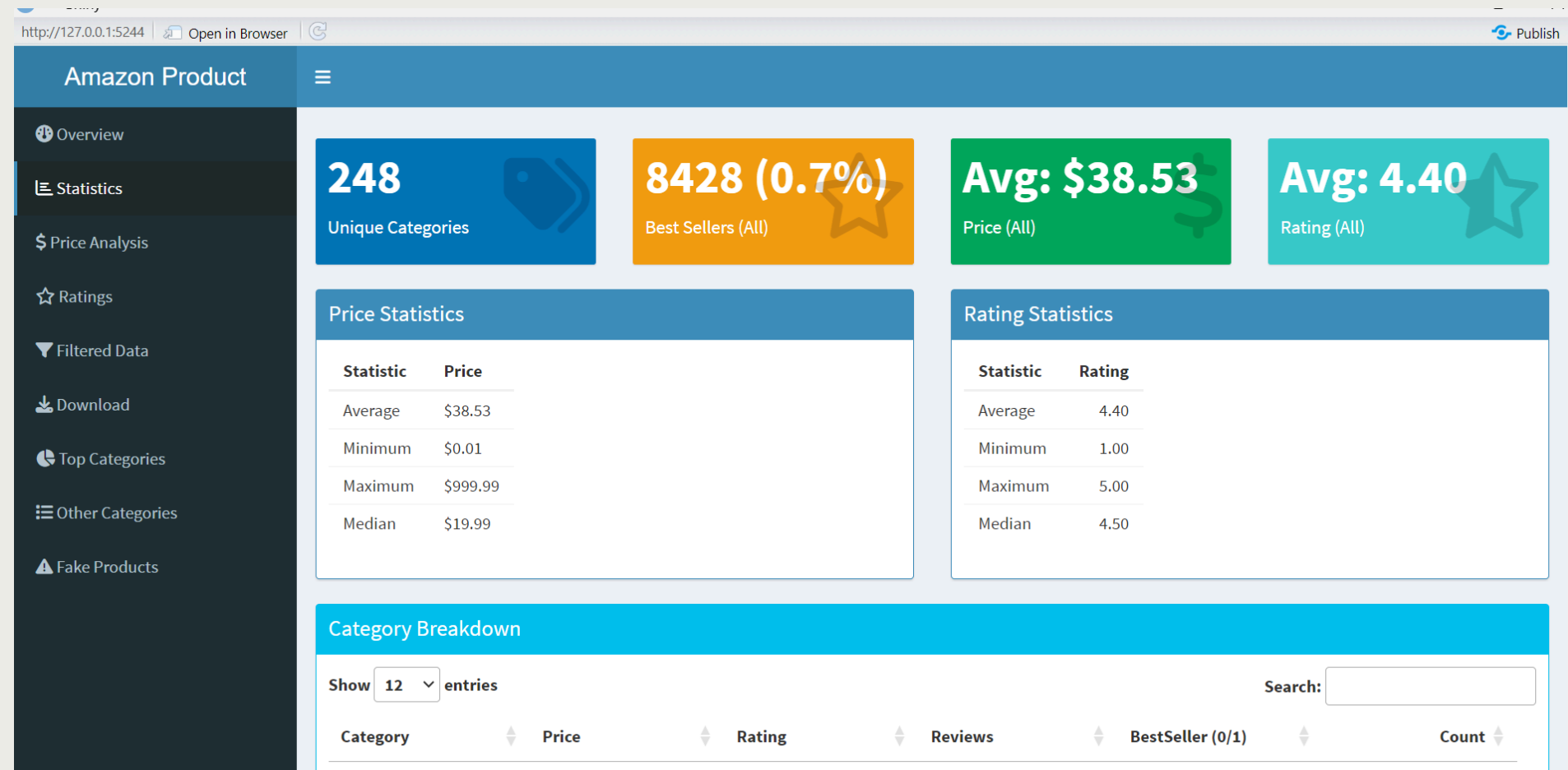
- Help e-commerce platforms visualize product data effectively
- Improve marketing and sales strategies
- Handle and explore large datasets (1.2M+ rows) efficiently
- Build skills in R, data visualization, and dashboard development

Project Objectives

- Extract meaningful insights from Amazon product data
- Enable interactive filtering and exploration
- Detect potentially fake products
- Provide user-friendly UI with data download and chart export features



ABOUT THE DATA

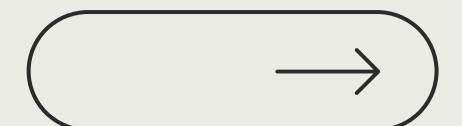


Dataset Used

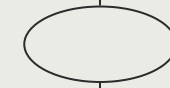
- amazon_products.csv – Titles, prices, ratings, reviews, bestseller status
- amazon_categories.csv – Category mapping
- Rows: ~1.2 million
- Source: [Amazon Products Dataset \(Kaggle\)](#).

Data Cleaning & Preprocessing

- Merged product and category data
- Removed duplicates and missing values
- Converted columns to proper data types
- Removed invalid prices/ratings
- Used keywords + logic to detect fake products



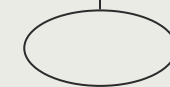
TOOLS & TECHNOLOGIES



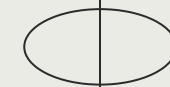
R PROGRAMMING



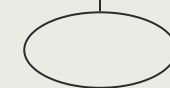
SHINY & SHINYDASHBOARD



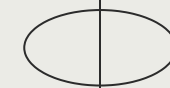
PLOTLY FOR INTERACTIVE PLOTS



DT FOR DATA TABLES



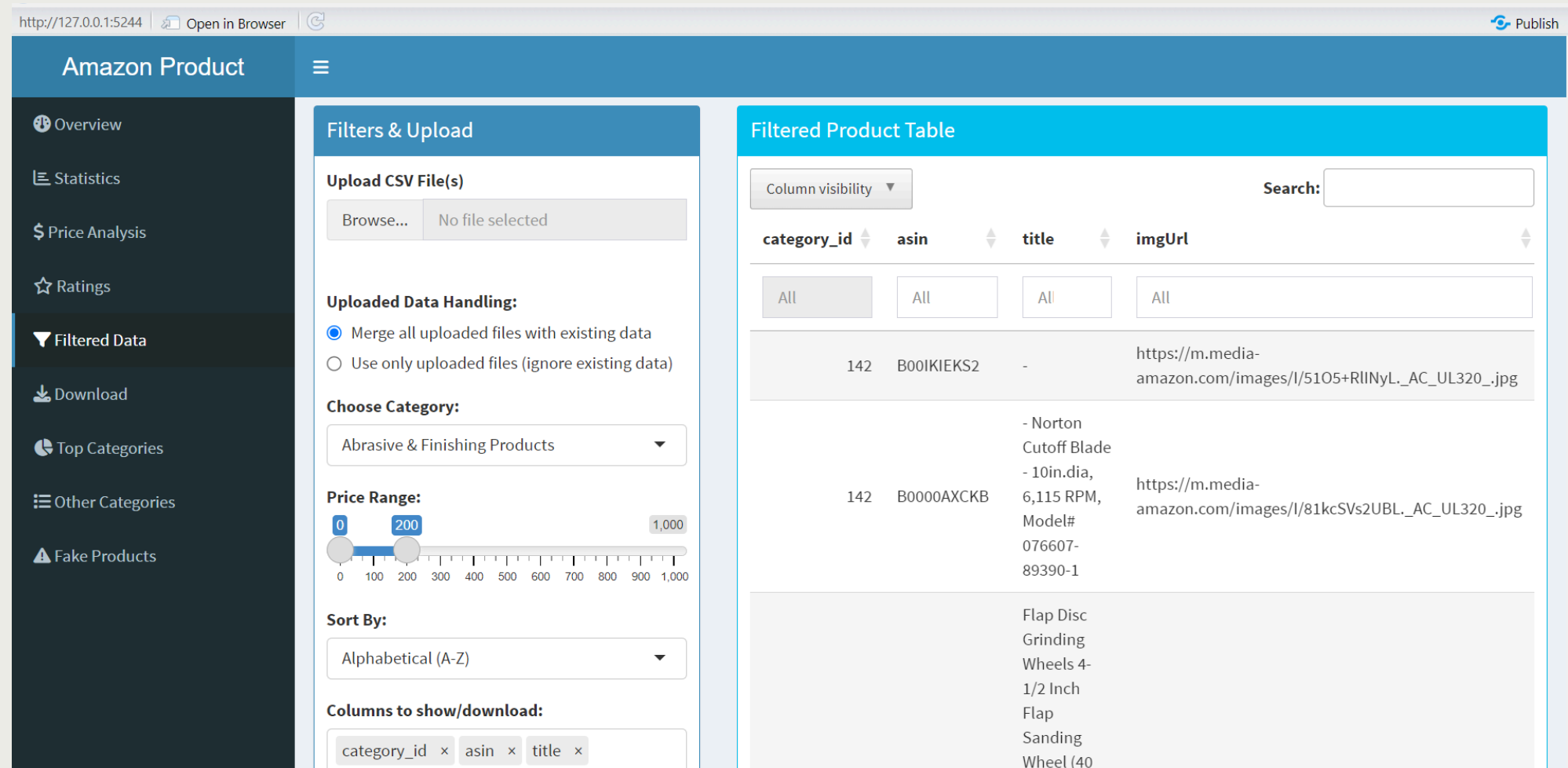
WEBSHOT2, PNG, TIFF – FOR PLOT EXPORT



**CUSTOM CSS + SHINYCSSLOADERS FOR
STYLING AND EFFECTS**



KEY FEATURES

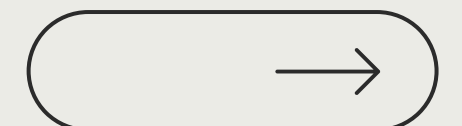


The Dashboard

- Overview Page with Data Summary
- Price & Ratings Distribution
- Filtered Data View + Download
- Top/Other Categories Visualization
- Fake Product Detection
- Trend Over Time & Brand Comparison
- User Data Upload & Merge

User Interface Design

- Sidebar navigation for quick access
- Responsive layout with boxes and fluid rows
- Sticky sidebar and header for better usability
- Interactive components with dynamic rendering



ACHIEVEMENTS & FUTURE SCOPE

Achievements

- Built a complete R Shiny web app
- Learned and applied reactive programming concepts
- Gained hands-on experience in data wrangling and visualization
- Improved skills in UI/UX design for interactive dashboards

Future Enhancements

- Incorporate ML models to detect fake reviews
- Support exporting high-quality plots (SVG/TIFF)
- Enable multilingual support and accessibility
- Optimize performance using caching and modular structure



MORE SCREENSHOTS



Amazon Product

Overview

Statistics

Price Analysis

Ratings

Filtered Data

Download

Top Categories

Other Categories

Fake Products

Filters & Upload

Upload CSV File(s)

Browse...

No file selected

Uploaded Data Handling:

☒ Merge all uploaded files with existing data

☐ Use only uploaded files (ignore existing data)

Choose Category:

Abrasive & Finishing Products

Price Range:

0

200

1,000

Sort By:

Alphabetical (A-Z)

Columns to show/download:

category_id

asin

title

Filtered Product Table

Column visibility

category_id

asin

title

imgUrl

productURL

stars

reviews

price

listPrice

isBestSeller

boughtInLastMonth

category_name

Search:

title	imgUrl
All	All
-	https://m.media-amazon.com/images/I/51O5+RIINyL._AC_UL320_.jpg
- Norton Cutoff Blade - 10in.dia, 6,115 RPM, Model# 076607-89390-1	https://m.media-amazon.com/images/I/81kcSVs2UBL._AC_UL320_.jpg
Flap Disc Grinding Wheels 4-1/2 Inch Flap Sanding Wheel (40	

These products are flagged based on heuristic signals like high rating but very few reviews, suspicious keywords, or being marked as best sellers with low purchase history.

•

title	category_name	price	stars	reviews	isBestSeller
Kandi Beads Bracelet Making Kit, Rainbow Pony B...	Beading & Jewelry Making	11.99	4.5	0	false
Multi-Fiber Stretch Jewelry Bead Cord .7mm 25 M...	Beading & Jewelry Making	11.49	4.6	0	false
BEADNOVA 0.6mm Bracelet String Clear Craft Wire...	Beading & Jewelry Making	5.99	4.5	0	false
26 Pcs 5 Inch Letter Mold Letter Resin Mold Alp...	Beading & Jewelry Making	44.99	5	4	false
QIANHAILIZZ 30 Silver Cap 2-1/4 Inch Faux Suede...	Beading & Jewelry Making	8.99	4.7	20	false
Hicarer 80 Pieces Word Charms Pendants Engraved...	Beading & Jewelry Making	9.86	4.6	0	false
Copper Wire 99.95% Pure – 16 Gauge 1.3mm – 42 F...	Beading & Jewelry Making	9.99	5	3	false
5Inch Long Needle Nose Pliers Jewelry Making Tools	Beading & Jewelry Making	6.97	4.6	0	false
NEW Silicone Craft Mat,20"x16"Silicone Art Mat ...	Beading & Jewelry Making	28.99	4.9	0	false
BigOtters Acrylic Keychain Blanks,120pcs Clear ...	Beading & Jewelry Making	6.95	4.6	0	false

Amazon Product

☰

👤 Overview

☰ Statistics

\$

Price Analysis

☆ Ratings

🔽 Filtered Data

📄 Download

📊 Top Categories

☰ Other Categories

⚠ Fake Products

Price Distribution

📷 🔍 ↕ 📐 💬 + - 🔗 🏠 ⬅ ➡ 📊

Price Range	Count
-100 - 100	1.175844M
100 - 200	~0.05M

Format:
PNG

Amazon Product

☰

- 📊 Overview
- 📋 Statistics
- 💰 Price Analysis
- ★ Ratings
- ▼ Filtered Data
- 📄 Download
- 📊 Top Categories
- ☰ Other Categories
- ⚠ Fake Products

Showing 1 to 10 of 6,759 entries

Previous

1

2

3

4

5

...

676

Next

